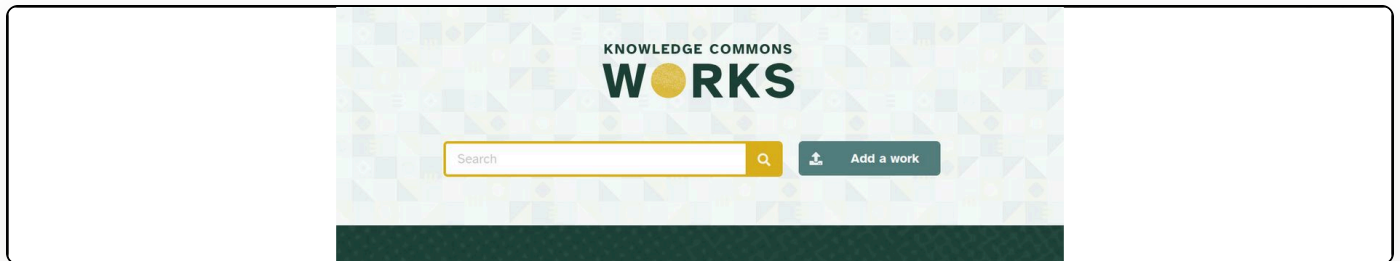


# Getting Knowledge Commons Works running locally

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**T**his week, I have started work at Michigan State University, as interim technical lead on the Knowledge Commons project. I'll probably say more about this at some point soon.

However, a huge part of onboarding at new digital/software places is getting your head around the stack and trying to get a development environment setup and ready to use. Often, this means relying on documentation that has fallen out of sync and/or trying to intuit things that some other developer thought was obvious (not blaming anyone, there, just noting it). So, in the spirit of documenting things for anyone else trying to do this, here's my notes on getting an instance of Knowledge Commons Works (the repository component) up and running locally.

First: the quick start instructions are fairly good.<sup>1</sup> However, what isn't clear is what you need in the .env file as a bare minimum. Mine looks like this:

```
INVENIO_INSTANCE_PATH=/opt/invenio/var/instance
INVENIO_RECORD_IMPORTER_LOCAL_DATA_DIR=/
INVENIO_RECORD_IMPORTER_DATA_DIR=/opt/invenio/var/import_data
INVENIO_SEARCH_DOMAIN='search:9200'
INVENIO_SITE_UI_URL="https://localhost"
INVENIO_SITE_API_URL="https://localhost/api"
REDIS_DOMAIN='cache:6379'
INVENIO_SQLALCHEMY_DATABASE_URI="postgresql+psycopg2://kcworks:PASSWORDHERE@db/kcworks"
# DON'T FORGET TO CHANGE THE PASSWORD HERE
POSTGRES_USER=kcworks
POSTGRES_DB=kcworks
INVENIO_CSRF_SECRET_SALT='GENERATE_IT_AS_PER_INSTRUCTIONS'
INVENIO_SECURITY_LOGIN_SALT='GENERATE_IT_AS_PER_INSTRUCTIONS'
INVENIO_SECRET_KEY='SECRET_KEY_VERY_SECRET'
COMMONS_API_TOKEN=mytoken # this must be obtained from the Commons administrators -
just leave as is
COMMONS_SEARCH_API_TOKEN=mytoken # this must be obtained from the Commons
administrators - just leave as is
INVENIO_DATACITE_PASSWORD=myinveniodatacitepassword # this must be obtained from the
Commons administrators - just leave as is
API_TOKEN=myapitoken # just leave as is
PYTHON_LOCAL_GIT_PACKAGES_PATH=/home/martin/Documents/Programming/MESH/knowledge-
commons-works # set this to the base directory
PYTHON_LOCAL_SITE_PACKAGES_PATH=~/.local/share/virtualenvs/knowledge-commons-works-
4DIkhDKF/lib/python3.12/site-packages # you need this for dev
POSTGRES_PASSWORD=PASSWORDHERE
PGADMIN_DEFAULT_EMAIL=martin@eve.gd
PGADMIN_DEFAULT_PASSWORD=PASSWORDHERE
INVENIO_LOCAL_INSTANCE_PATH=/opt/invenio/var/instance
```

In order to populate the virtualenv, I had to make some changes to the Pipfile so that it would run on my machine:

```
[[source]]
name = "pypi"
url = "https://pypi.org/simple"
verify_ssl = true
[dev-packages]
check-manifest = ">=0.25"
halo = ""
jsonlines = ""
requests-mock = ""
selenium = ""
docker-services-cli = ""
[packages]
aiohttp = ""
async-timeout = ">=4.0.3"
cchardet = ""
flask-admin = "==1.6.1"
flask-babel = ""
flask-breadcrumbs = ""
flask-principal = ""
greenlet = {}
halo = ""
invenio-app-rdm = { extras = ["opensearch2"]}
invenio-communities = { editable = true, path = "./site/kcworks/dependencies/invenio-communities" }
invenio-group-collections-kcworks = { editable = true, path =
"./site/kcworks/dependencies/invenio-group-collections-kcworks" }
invenio-modular-deposit-form = { editable = true, path =
"./site/kcworks/dependencies/invenio-modular-deposit-form" }
invenio-modular-detail-page = { editable = true, path =
"./site/kcworks/dependencies/invenio-modular-detail-page" }
invenio-rdm-records = { editable = true, path = "./site/kcworks/dependencies/invenio-rdm-records" }
invenio-record-importer-kcworks = { editable = true, path =
"./site/kcworks/dependencies/invenio-record-importer-kcworks" }
invenio-records-resources = { editable = true, path =
"./site/kcworks/dependencies/invenio-records-resources" }
invenio-remote-api-provisioner = { editable = true, path =
"./site/kcworks/dependencies/invenio-remote-api-provisioner" }
invenio-remote-user-data-kcworks = { editable = true, path =
"./site/kcworks/dependencies/invenio-remote-user-data-kcworks" }
invenio-s3 = ""
invenio-saml = ""
invenio-subjects-fast = ""
invenio-utilities-tuw = ""
invenio-vocabularies = { editable = true, path = "./site/kcworks/dependencies/invenio-vocabularies" }
isbnlib = ""
```

```
langdetect = ""
numpy = ""
pip = ""
pytest-invenio = ""
python-dotenv = ""
python-iso639 = ""
python-stdnum = ""
selenium = ""
sqlalchemy = { extras = ["asyncio"]}
timefhuman = ""
titlecase = ""
tqdm = ""
unidecode = ""
uwsgi = ">=2.0"
uwsgi-tools = ">=1.1.1"
uwsgitop = ">=0.11"
xmlsec = "<1.3.14"
kcworks = {file = "site", editable = true}
[requires]
python_version = "3.9"
[pipevn]
allow_prereleases = true
```

The main change here was:

```
cchardet = ""
```

This allowed me to run:

```
pipenv install --dev --python=3.12
```

Then you need to:

```
docker-compose --file docker-compose.yml up -d
```

and

```
docker exec -it kcworks-ui bash
```

Once you are in the container, you should be able to run:

```
bash ./scripts/setup-services.sh
```

and select local storage, which will get everything setup. If you get database access errors, take a close look at the .env file and ensure that the INVENIO\_SQLALCHEMY\_DATABASE\_URI has the password set. I didn't realise for quite some time that you have to update the schema here.

1. See <https://github.com/MESH-Research/knowledge-commons-works> ↩